

Implement Network Security Defenses

- Device Hardening
 - Disable Unused Ports
 - ```
enable
conf t
int fa 0/1
shut
```
  - Change Default Passwords
    - <https://cirt.net/passwords>
- NAC
  - Port Security
  - 802.1X
    - Supplicant
      - Extensible Authentication Protocol(EAP)
    - Authenticator
      - Connects the Supplicant to the Authentication Server
    - Authentication Server
      - Receives and Responds to requests for network access
        - Typically running RADIUS and EAP
  - MAC Filtering
- Key Management
- Security Rules
  - ACLs
  - URL Filtering
    - Known Malicious Websites
    - Adult/Gambling Websites
  - Content Filtering
    - Categories
      - News
      - Social Media
      - Adult
      - Gaming
      - File Types
  - Vendors
    - <https://www.barracuda.com/products/network-protection/web-security-gateway>
    - <https://umbrella.cisco.com/>
- Zones
  - Trusted vs. Untrusted
  - Screened Subnet